

Richael

Auckland, New Zealand | huangd6666@gmail.com
[dh8116.github.io](https://github.com/dh8116) | github.com/dh8116 | x.com/RicaV42

PROFILE

A student building GPU kernels, fine-tuned language models, and full-stack applications with real users. Nine Triton kernels benchmarked against PyTorch over nine consecutive weeks, each published with its numbers; a fine-tuned Qwen3-14B serving in production behind vLLM, leading the provider chain and answering faster than the model it replaced.

EDUCATION

Year 11, Auckland, New Zealand. Expected graduation 2028

Relevant coursework: IB Math AA HL, NCEA level 1 (planning for AP calculus & cs exams next year)

TECHNICAL SKILLS

GPU & ML: Triton, PyTorch, Python, GPU benchmarking and profiling, LoRA fine-tuning, vLLM, Modal, eval harnesses

Full-stack: TypeScript, Node.js, Express, Next.js, React, Supabase/Postgres, Web Audio API, Stripe

Practice: Git, GitHub Actions, Playwright end-to-end testing, web security hardening (CSP, XSS)

GPU SYSTEMS & OPTIMIZATION

One Triton kernel per week, benchmarked against PyTorch.

- Cut fused cross-entropy to 15.90 ms against PyTorch's 24.03 ms at vocab 131,072 (1.51x), at 1.67x less peak memory, by collapsing five live copies of the logits tensor to three.
- Raised fused softmax from ~55 GB/s to ~230 GB/s, and established T4's ~240 GB/s practical ceiling by measurement.

DISTRIBUTED AI INFRASTRUCTURE & DISPATCH

- Fine-tuned Qwen3-14B with LoRA on a 1,526-row corpus spanning five generation paths in six languages; achieving full performance for \$1.30 of compute.
- Deployed optimized cold-start vLLM architecture, leading the production chain at 2.3–4.1s against the original 5.7s model.
- Built warm-gate routing: the gateway probes for a warm container and the probe doubles as the warm-up, so a 108 s cold start never reaches a user.
- Fine-tuned MusicGen-small for bar-locked stems, built the eval harness scoring clips as production.

PROJECTS

Soulor AI — soulor-ai.vercel.app

- A regional-variant companion and multi-perspective simulation app with cross-session memories and full relationship stages. Also serving as a great tool for daily reminders and wellness enhancement.
- Cross-Mode Engine | Relationship Staging | Bias Adaptation | Stateful Sampling

VNportal — vnportal.vercel.app

- A vinyl and music community platform for creation, recreation, and discussion, promoting innovative music creation with AI-assistance.
- Controllable Audio Synthesis | Onset Charting | Source Cascade | Stem Synthesis

Personal website — dh8116.github.io

- A central hub sharing my skills, projects, current development stages, and blog writings.
- Next.js static export | Hreflang Sitemap | Postbuild CSP.

RESEARCH & WRITING

- Leading a five-person-team paper on RLVR verifier persistence: when a persistently wrong verifier harms GRPO training. Targeting COLM/TMLR.
- Publishing general and kernel writeups at dh8116.github.io/blog; co-authored several modelling papers for HiMCM and IMMC.

AWARDS & VOLUNTEERING

- Score 10/15, AIME (American Invitational Mathematics Examination) | Honor Roll (Top 5%), AMC 12 (The American Mathematics Competition 12) | High Distinction, AUSAMC (Australian Mathematics Competition)
- Honorable Mention, HiMCM (High School Mathematical Contest in Modeling)
- Honorable Mention, IMMC regional (International Mathematical Modeling Challenge Regional Round)
- Volunteer, Mustang Math Community — Technology and Curriculum Development

黄陶朵儿

中国上海 | huangd6666@gmail.com | 18217529014 / +64 2902575400

[dh8116.github.io](https://github.com/dh8116) | github.com/dh8116 | x.com/RicaV42

个人简介

11年级学生, 正在开发GPU内核、微调语言模型以及面向真实用户的全栈应用程序。连续八周对八个Triton内核与PyTorch进行了性能基准测试, 并公布了每个内核的测试数据; 一个微调后的Qwen3-14B模型已在vLLM后端投入生产环境, 在供应链中处于领先地位, 响应速度比其替换的模型更快。

教育背景

新西兰St Cuthbert's College (奥克兰圣卡斯伯特学院) — 11年级, 预计2028年12月毕业

相关课程: IB Math AA HL, NCEA level 1 (计划明年参加AP calculus和计算机考试)

技术技能

GPU 与机器学习: Triton、PyTorch、Python、GPU 基准测试与性能分析、LoRA 微调、vLLM、Modal、评估框架

全栈开发: TypeScript、Node.js、Express、Next.js、React、Supabase/Postgres、Web Audio API、Stripe

实践: Git、GitHub Actions、Playwright 端到端测试、Web 安全加固(CSP、XSS)

GPU 系统与优化

每周实现一个 Triton 内核, 并与 PyTorch 进行性能对比。

- 通过将对数几率张量的五个活动副本合并为三个, 在词汇量为 131,072 时, 将融合交叉熵计算时间从 PyTorch 的 24.03 毫秒缩短至 15.90 毫秒 (1.51 倍), 同时峰值内存使用量减少了 1.67 倍。
- 将融合软最大值 (softmax) 的吞吐量从约 55 GB/s 提升至约 230 GB/s, 并通过实测确立了 T4 约 240 GB/s 的实际性能上限。

分布式 AI 基础设施与调度

- 在包含六种语言、五条生成路径的 1,526 行语料库上, 利用 LoRA 对 Qwen3-14B 进行微调; 仅需 1.30 美元的计算成本即可达到全性能水平。
- 部署了优化的冷启动 vLLM 架构, 生产链响应时间从原始模型的 5.7 秒缩短至 2.3–4.1 秒, 处于行业领先地位。
- 构建了“暖门”路由机制: 网关会探测“暖”容器, 且该探测过程本身即作为预热过程, 因此长达 108 秒的冷启动过程绝不会影响用户。
- 针对小节锁定的音轨对 MusicGen-small 进行了微调, 并构建了评估框架, 对音频片段进行与生产环境相同的评分。

项目

Soulor AI — soulor-ai.vercel.app

- 一款具备跨会话记忆和完整关系阶段的、具有地域变体功能的伴侣及多视角模拟应用。同时也是日常提醒和健康管理
- 的绝佳工具。
- 跨模式引擎 | 关系阶段 | 偏好适应 | 带状态采样

VNportal — vnportal.vercel.app

- 一个致力于创作、娱乐和讨论的黑胶唱片与音乐社区平台, 旨在推动借助 AI 辅助的创新音乐创作。
- 可控音频合成 | 起音图谱 | 音源级联 | 音轨合成

个人网站 — dh8116.github.io

- 一个集中展示我的技能、项目、当前开发阶段及博客文章的枢纽。
- Next.js 静态导出 | Hreflang 网站地图 | 构建后 CSP。

研究与写作

- 领导一个五人团队撰写关于 RLVR 验证器持久性的论文: 当一个持续出错的验证器损害 GRPO 训练时。目标为 COLM/TMLR 会议。
- 在 dh8116.github.io/blog 发布通用及内核相关的技术报告; 合著了多篇针对 HiMCM 和 IMMC 的建模论文。

奖项

- AIME (美国邀请数学考试) 得分 10/15 | AMC 12 (美国数学竞赛 12) 荣誉榜 (前 5%) | AUSAMC (澳大利亚数学竞赛) 最高荣誉
- HiMCM (高中数学建模竞赛) 荣誉提名 | IMMC 区域赛 (国际数学建模挑战赛区域赛) 荣誉奖